

FORENSIC ENGINEER'S REPORT

Halberd Forensic Engineering · Our reference HFE-2025-1188

INSTRUCTIONS

Instructed by:	Ridgeway Kessler Brokerage Services
On behalf of:	Fairmount Textile Mills Holdings, Inc.
Policy number:	CP-2210-55870
Author:	P. Vanterpool-Aziz, PE, CFEI
Instructed by:	Allegheny Commercial Risk Company
Instructed on:	15 August 2025
Site attendance:	16 August 2025
Scope:	Origin and cause, and the preservation of evidence

AFFECTED PLANT AND BUILDINGS

Jet Dye Machine No. 4:	Total loss
Jet Dye Machines 3 and 5:	Heat distortion, smoke and water
Building 1B:	18,000 sq ft fire, smoke and sprinkler water
MCC-1B-2 and PLC cabinets:	Destroyed
Building 1A east:	Soot deposition over 20,000 sq ft

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Circumstances and findings

CIRCUMSTANCES

Circumstances:

Building 1B houses six jet dyeing machines on a common thermal oil ring main. Machine No. 4 underwent a planned hydraulic hose replacement on 11 August. The machine ran two shifts on 12 August without incident and failed on the second shift of 13 August.

FINDINGS

Findings:

The origin is at the hydraulic power unit on the drive end of Machine No. 4, approximately 600 mm above the thermal oil supply line. The replaced hose has separated at the crimped ferrule. The ferrule bore shows no crimp witness marks over roughly a third of its circumference, consistent with an undersized or incorrectly seated crimp rather than with in-service fatigue. Thermal oil supply temperature at the time was logged at 271 degrees C, above the 230 degrees C auto-ignition temperature of the hydraulic fluid in use. The failed hose assembly, the ferrule and the crimp die records have been secured.

Date of loss:	13 August 2025
Cause of loss:	Fire — hydraulic fluid release onto a hot thermal oil line
Incident reference:	GCFD-25-0081144
Injuries reported:	2

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Preliminary quantum and reserve advice

PRELIMINARY QUANTUM

Basis:

The figures below are preliminary and are given for reserving purposes only. They are subject to verification and may move as the investigation continues.

Jet Dye Machine No. 4 replacement:	USD 4,100,000
Machines 3 and 5 refurbishment:	USD 2,300,000
MCC, PLC and cable tray:	USD 800,000
Building 1B structure, roof and MEP:	USD 4,800,000
Baghouse and ductwork:	USD 340,000
Smoke remediation, Buildings 1A and 1B:	USD 1,600,000
Estimated loss:	30,500,000
Currency:	USD

ON THE LINT REMOVAL WARRANTY

Comment:

The policy warrants documented weekly lint removal from the weave room and the dust collection ductwork. Cleaning logs were produced for Building 1A through 8 August, but the Building 1B finishing-line duct logs record no cleaning after 19 July, a gap of twenty-five days. Duct interior photography shows accumulation consistent with that interval. The lint did not cause the fire; it materially extended it. This is recorded as a fact for the coverage file, not as an opinion on cover.

RECOMMENDATION

Recommendation:

Preserve the hose assembly, the crimp records and the maintenance work order. Notify the hose supplier and Upstate Industrial Maintenance, LLC, who performed the 11 August replacement, before any destructive testing.